

ONLINE MASTERS IN **DATA SCIENCE**

DSC 207 - PYTHON FOR DATA SCIENCE

ARRAYS IN NUMPY

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By the end of this video, you will be able to:

By the end of this lecture, you should be able to:

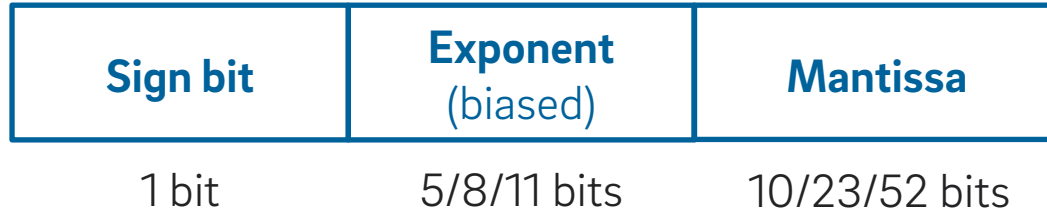
- Describe the role of data types in storing NumPy arrays
- How to visualize NumPy arrays



Common Data Types

Data Type	Explanation
Bool	True or False
int{8/16/32/64}	Signed integers stored using a particular number (n) of bits. n is the value after the "int" Range is $-2^{(n-1)}$ to $2^{(n-1)}-1$. For example, int16 can store values between -32,768 to 32,767
uint{8,16,32,64}	Unsigned integers stored using a particular number (n) of bits. Range is between 0 to 2^n-1 . For example, uint8 stores values between 0 to 255.
float{16/32/64}	Signed decimal values stored using Half precision (16 bits), Single precision (32 bits), and Double precision (64 bits). More bits allow for a larger range of values to be represented.

Floating Point Numbers



Example: Represent +29.5 in base 2 scientific notation:

$$+29.5_{10} = +11101.1_2 = +1.11011 \cdot 2^4$$

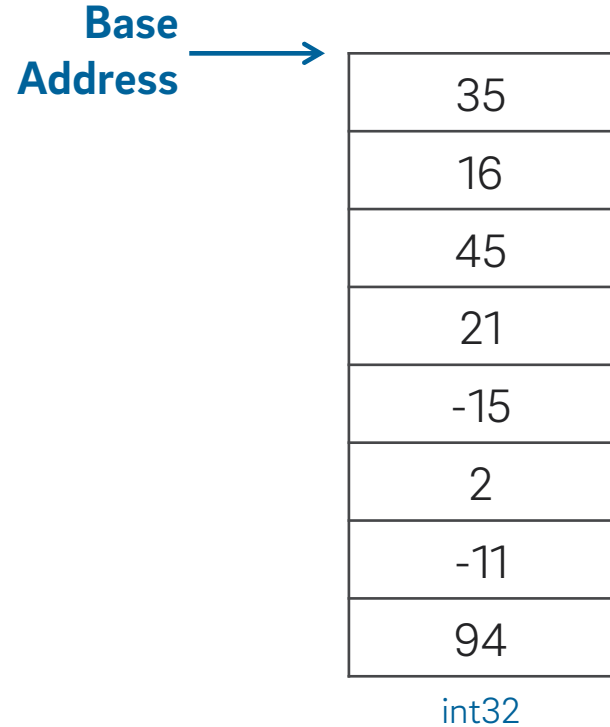
Diagram illustrating the components of the base 2 scientific notation:

- Sign bit:** The '+' sign.
- Mantissa:** The fractional part of the normalized number, $.11011$.
- Exponent:** The power of 2, 2^4 .

Why Type Matters

Speed!

- Computers store data in memory at "byte addresses"
- If you have an array (named "arr") of int32, each integer takes 4 bytes.
- arr[0] is at the base address
- arr[4] is at the base address+16
- This makes it really easy, and fast, to access any value in memory



This is why NumPy arrays need 1 type

How about these values to the right?

- You can't store a string like "How are you?" in 32 bits.
- So, if you try to store data that has more than one type, it will become a python list rather than a NumPy array
 - This is even true if the types are the same size, e.g., int32 and float32

Base Address →

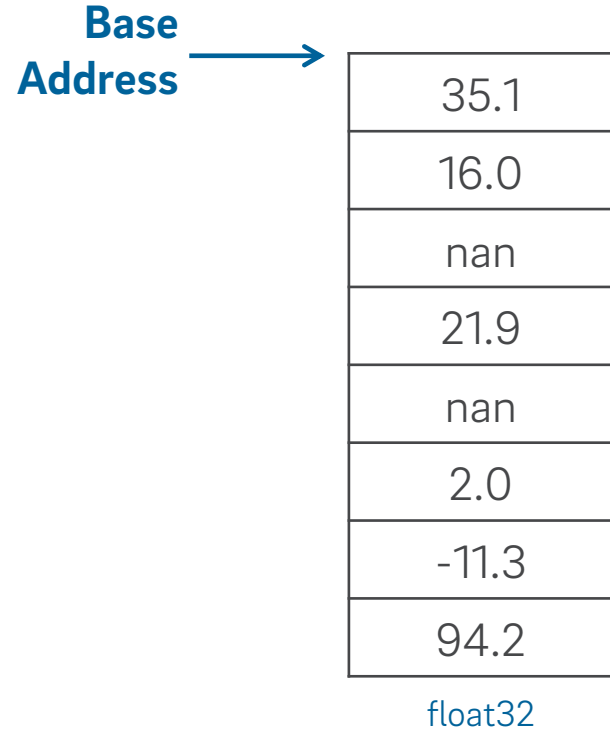
35
"How are you?"
-75.2
21
-15
2
-11
94

What data type?

Missing data (for floating point arrays only)

Np.nan

- Used to show missing data
- Limited to floats
- Look for special functions that handle nans:
 - `isnan(array)`
 - `nansum(array)`
 - `nanmin(array)`



Multiple dimensions!

- NumPy arrays can be multi-dimensional!
- They are actually still stored in memory in a 1-D array (as memory is a 1-D array, in a sense)
 - Multiple dimension arrays are accomplished through just complex indexing

